

**Time:** 5 minutes**Outcome:** Connect Market Structure, Price Taking & Firm Revenue.**Difficulty:** ●●○ Intermediate

Market Structure, Price Taking & Firm Revenue



THE CORE IDEA

A perfectly competitive firm takes market price (P) as given. Each additional unit adds the same amount to revenue without lowering the price on previous units. Thus average revenue (AR) and marginal revenue (MR) both equal price: $P = AR = MR$.



HOW TO RECOGNIZE IT

- Look for many firms, a standardized product, easy entry, and no control over market price.
- A price-taking firm can sell another unit without lowering its price.
- For the competitive firm, price, average revenue, and marginal revenue are equal.
- Treat market price as a horizontal line in the individual firm's graph.



WATCH OUT

A common mistake is to separate price, average revenue, and marginal revenue for a price-taking firm. Under perfect competition, $P = AR = MR$.



WORKED EXAMPLE: REVENUE FOR A PRICE TAKER

Price-taking firm at a market price of \$25

	Price	Total revenue	MR per extra unit
$Q = 48$	\$25	\$1,200	\$25
$Q = 49$	\$25	\$1,225	\$25
$Q = 50$	\$25	\$1,250	\$25

The firm takes a market price of \$25. Selling 50 units yields total revenue of $\$25 \times 50 = \$1,250$. Each additional unit adds \$25 to revenue without lowering the price on earlier units. Thus $P = AR = MR$. This revenue relationship does not by itself identify the best output; costs also matter.



CHECK YOURSELF

Why does total revenue rise linearly with output?

**READY?**

Return to the game and master this concept.